ORM-RP Form 01
Revised 2023

PRESCRIBED FORMAT PROJECT PROPOSALS

(1) PROJECT PROFILE

Program Title:

Project Title **: eTee: Manure Management for Small-Scale Chicken Coops using IoT**

Designation	Name	College	RELE
Project Leader			
Project Members	Tristan E. Listanco	CCS	
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(***Title****- the identification of the Program and the component projects.*
Project*- refers to the basic unit in the investigation of specific S&T problem/s with predetermined objective/s to be accomplished within a specific time frame.*
Project Leader*- refers to the person who plans, organizes and supervises the overall activities of a project.*
Project Member/s*-refers to the person who is part of the research team.*
RELE: *Research and Extension Load Equivalence, please refer to the BOR for RELE)*

(2) TYPE OF RESEARCH

_____ Basic
__✓__ Applied

__✓__ New
_____ Continuing

(***Type of Research****- indicates whether the project is basic or applied.*
Basic research*- is an experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundations of phenomena and observable facts, without any particular or specific application or use in view.*
Applied research*- is an original investigation undertaken in order to acquire new knowledge directed primarily towards a specific aim or objective*.)

(3) R&D PRIORITY AREA & PROGRAM (based on HNRDA 2017-2022)

__✓__ Agriculture, Aquatic and Natural Resources
Sector: _____

_____ Health
Sector: _____

_____ Industry, Energy and Emerging Technology
Sector: _____

_____ Disaster Risk Reduction and Climate Change Adaptation
Sector: _____

_____ Basic Research
Sector: _____

(Indicate in which R&D agenda the project can be categorized. Indicate also the specific Sector, whether crops, livestock, forestry, agricultural resources or socio-economics; fisheries or aquatic resources; biotechnical, pharmaceutical, or health services; biotechnology, information technology, material science, photonics or space technology; industry, energy, utilities or infrastructure, etc.)

R&D Priority Area & Program (based on MSU-IIT Integrative Research Clusters)

_____ 1 Culture, Technology, and Society
_____ 2 Peace and Governance

R&D Priority Area & Program (based on UNDP 17 SDGs)

_____ 1 No Poverty
_____ 2 Zero Hunger
__✓__ 3 Good Health & Well-Being

_____ 3 Integrative Resiliency and Sustainable Solutions	_____ 4 Quality Education
___✓___ 4 Disruptive innovation and smart nation	_____ 5 Gender Equality
_____ 5 Applied and theoretical science	_____ 6 Clean Water & Sanitation
_____ 6 Teaching and Learning	_____ 7 Affordable & Clean Energy
_____ 7 Natural Products and Biotechnology	_____ 8 Decent Work & Economic Growth
_____ 8 Financial and Risk Management	___✓___ 9 Industry, Innovation & Infrastructure
_____ 9 Materials Science	_____ 10 Reduced Inequalities
	___✓___ 11 Sustainable Cities & Communities
	___✓___ 12 Responsible Consumption & Production
	_____ 13 Climate Action
	_____ 14 Life Below Water
	_____ 15 Life on Land
	_____ 16 Peace, Justice & Strong Institutions
	___✓___ 17 Partnership for the Goals
	(Indicate in which R&D agenda the project can be categorized.)

(4) RATIONALE

The livestock industry represents a crucial sector in the national economy, accounting for nearly 4% of global GDP, with about 1.4% attributed to poultry production. Poultry farming plays a particularly significant role in developing countries where it serves as a primary source of livelihood and nutritional value (Mottet & Tempio, 2017; Okubanjo et al., 2025). In the Philippines, the poultry industry has seen substantial growth and serves as a backbone of the country's food system. However, this expansion has created significant waste management challenges, particularly regarding chicken manure. The country produces an estimated 76 million metric tons of chicken manure annually, with improper disposal practices leading to severe environmental degradation and public health threats (Parihar et al., 2019). Adult chickens excrete between 150-160 grams of droppings daily, contributing to the substantial waste burden (Ngwabie et al., 2018). Improper waste handling has been linked to respiratory and digestive problems among individuals living near poultry farms, with studies showing that 60% of neighbors reported respiratory issues, while 40% experienced digestive problems due to exposure to poultry waste. Current chicken manure disposal practices in the Philippines vary widely, Disposal in closed or open pits, Improper discharge into water bodies including rivers and canals leaving manure on the ground to decompose (7.3 Manure Disposal for Survey Households, 2024)

Several studies have emerged in poultry farming that utilize temperature and humidity sensors (DHT11), WiFi and Bluetooth enabled microcontrollers (ESP8266) for data transmission, and real-time database storage solutions like Google Firebase (Widayanto & Aji, 2025). Other applications include using MQ135 sensors to detect Carbon Dioxide and implementing DC Fan Systems to aid proper ventilation inside chicken coops (Lifwarda et al., 2025). Other studies also demonstrate the use of machine learning models in computer vision to detect fecal information using smartphones. For instance, YOLO-V3 was employed to segment regions of interest (ROI) from fecal images, while ResNet50 was utilized for classifying the segmented image into health conditions (Degu & Simegn, 2023). However,**no existing solution** integrates these technologies to solve the multiple difficulties in waste management practices, especially in Iligan City where several residents in sub-urban and upland areas maintain small-scale backyard chicken coops for everyday living and business.

This project aims to address the pressing issue of odor control and waste management in poultry farms by developing a technology-driven solution focusing on two key areas, the design and implementation of an, **embedded system for automatic odor control** using microcontrollers and environmental sensors and the development of an **IoT-based monitoring system** that allows remote tracking and analysis of key environmental data The solution will implement solar-powered automated systems to manage manure, including a conveyor belt, eco-friendly design, and harmful gas detection. It will also introduce and integrate digital sensors such as DHT11 (for temperature and humidity) and ammonia and carbon dioxide sensors as monitoring parameters.

SIGNIFICANCE:

The Local Small-Scale Chicken Coop Farmers

The development of an IoT-based smart chicken coop with automated manure management system holds significant value for local small-scale farmers who currently face numerous challenges in managing their poultry operations. It reduces labor and time investment by automating manure collection and processing thus significantly decreasing the manual labor required, allowing farmers to focus on other productive activities. The system enables proper processing of chicken manure,

converting a waste problem into a potential fertilizer product that can be sold to local crop farmers, aligning with sustainable practices where contract farmers tend to sell manure more frequently than independent operators

The Local Government Unit (LGU)

LGUs can support local economic development without compromising environmental standards, particularly important in regions like Northern Mindanao, which is among the leading poultry-producing areas in the country (Dili et al., 2022).The IoT system's data collection capabilities provide valuable information for LGUs to develop more effective and targeted policies regarding poultry farming and waste management. It also helps prevent contamination of local water bodies and soil, addressing the environmental degradation caused by improper disposal practices that contribute to the estimated 76 million metric tons of chicken manure produced annually in the Philippines (Parihar et al., 2019).

The Researchers

This project contributes to the growing body of knowledge on IoT applications in agriculture, supporting the United Nations' call for investing in smart agriculture, which includes food shortage mitigation, provision of nutrition for all by 2030, and enhancing food security. It creates opportunities for collaboration across engineering, agricultural science, environmental science, and public health domains, addressing cross-cutting challenges like the need to effectively manage and dispose of poultry manure. It also serves as a practical demonstration of sustainable farming technologies for agricultural education and extension programs, addressing the lack of awareness

(***Rationale****- discuss here the significance of the project to the advancement of knowledge and impact to the society. Not to exceed one (1) page.*)

(5) REVIEW OF LITERATURE

a) related researches that have been conducted

Chicken Coop Management Systems and Manure Control

Yunos et al. (2024) conducted a study on probiotic chicken products, focusing on ammonia-free poultry farming and food processing without chemicals, antibiotics, vaccines, or hormones. Their research emphasized the development of eco-smart, odorless poultry coop prototypes to attract new breeders to the field. The study demonstrated that probiotic methods using charcoal and sawdust effectively prevent odor in chicken coops by absorbing moisture. These findings align with earlier research by Shao et al. (2015), which confirmed that sawdust can significantly improve environmental conditions in poultry operations.

Complementing this approach, Wu et al. (2024) assessed optimal airflow inside chicken coops using Computational Fluid Dynamics (CFD) simulations. Their findings revealed that internal coop layout significantly impacts chicken welfare, particularly when wet curtains are closed. The arrangement of chicken racks within the coop was found to profoundly affect airflow organization, highlighting the importance of proper ventilation design in managing ammonia levels and moisture content.

Wireless Sensing Networks in Poultry Management

The health and production of chickens are influenced by environmental conditions including humidity, temperature, light, and ammonia gas. Studies show that the adoption rate of Precision Livestock Farming (PLF) technologies remains below 20% in most regions, primarily due to infrastructure limitations, lack of technical expertise, and limited access to affordable technology (Arago et al., 2022).

Widayanto and Aji (2025) integrated hardware and software components to create a comprehensive monitoring system for poultry environments. Their system utilized DHT11 sensors for measuring temperature and humidity, ESP8266 microcontrollers for data transmission, and actuators for maintaining air circulation. The software component involved the development of an Android application using Android Studio and Firebase for real-time data storage and retrieval, providing features such as graphical data visualization, historical data storage, and user authentication. Similarly, Sitaram et al. (2018) utilized a comparable setup but enhanced it with RTC Clock for real-time date and time logging and captured exhausted methane gas, converting it to electrical energy stored in batteries. T

Temperature regulation systems in chicken coops have shown substantial benefits in production outcomes. Hanau et al. (2025) reported that monitored Label Rouge free-range chickens exhibited an average weight 25% higher than unmonitored counterparts during the fifth week of evaluation.

Computer Vision Applications in Poultry Farming

Expert and intelligent systems based on computer vision algorithms are becoming a common part of agricultural production management. Agricultural automation systems experience improved economic performance, general performance, coordination, and robustness when computer vision technology is combined with artificial intelligence algorithms (Tian et al., 2020).

Visible-light digital cameras offer a cost-effective solution for monitoring poultry operations, with top-view camera imaging being least disturbing for animals while producing the most useful data (Kashiha et al., n.d.; Abd Aziz et al., 2021). Image analysis typically involves pre-processing, segmentation, feature extraction, and classification or prediction. Convolutional Neural Networks (CNN) have shown promising results in poultry health monitoring. When compared to Single Shot MultiBox Detector (SSD) and Improved Feature Fusion Single Shot MultiBox Detector (IFSSD), the latter demonstrated more accurate classification ability for health status, achieving 99.7% accuracy in detecting sick broilers compared to SSD's 98.7% accuracy (Zhuang & Zhang, 2019; Abd Aziz et al., 2021).

Wang et al. (2019) proposed an automated broiler digestive disease detection using deep CNN models to classify abnormal broiler droppings based on factors like shape, color, and water content. Their comparative study of Faster R-CNN and YOLO-V3 networks showed that Faster R-CNN achieved 99.1% recall and 93.3% mean average precision, while YOLO-V3 achieved 88.7% recall and 84.3% mean average precision. These findings are particularly relevant for manure management, as they enable early detection of health issues through fecal analysis. Degu and Simegn (2023) further advanced this field by developing a smartphone-based system that uses YOLO-V3 to segment regions of interest from fecal images, while ResNet50 classifies the segmented image into four health conditions: Health, Coccidiosis, Salmonella, and New Castle Disease.

(b) scientific basis/theoretical framework;

IoT in Agriculture Industry

The theoretical foundation of smart agriculture rests on the integration of Internet of Things (IoT) technologies to transform traditional farming into data-driven, efficient, and sustainable endeavors. In poultry farming, environmental conditions significantly influence chicken health and production. These critical parameters include humidity, temperature, light intensity, and ammonia gas concentration, which traditionally have been controlled manually (Sitaram et al., 2018).

The conceptual framework of Precision Livestock Farming (PLF) utilizes sensors, actuators, and data analytics to optimize production while simultaneously addressing environmental concerns. Despite the potential benefits, studies indicate that the adoption rate of PLF technologies remains below 20% in most regions, primarily due to infrastructure limitations, lack of technical expertise, and limited access to affordable technology (Arago et al., 2022). Regional implementation of IoT solutions has demonstrated measurable improvements in poultry operations. In Central Luzon, which hosts the largest concentration of poultry farms in the Philippines, IoT-based automated feeding systems have shown the potential to reduce feed wastage by up to 30% compared to manual methods (Ramizares et al., 2024). Similarly, Geographic Information System (GIS)-based monitoring approaches in Laguna province have enhanced farm management by providing data-driven insights into environmental impacts and resource allocation (Alcantara et al., 2009).

Progressive Web Apps in Internet of Things

The theoretical framework for system implementation incorporates Progressive Web Application (PWA) architecture, which bridges the gap between web applications and native mobile applications. PWAs offer advantages including offline functionality, cross-platform compatibility, and reduced resource requirements compared to traditional mobile applications. Mena et al. (2019) established a theoretical model for integrating geospatial data and IoT information within a PWA framework using microservices architecture. This approach addresses the limitation of IoT data typically being confined to monitoring dashboards without integration with other relevant datasets. The microservices architecture provides modularity, scalability, and maintainability advantages over monolithic application structures.

The component-based architecture proposed by Mena et al. (2019) utilizes a primary microservice responsible for generating component configurations based on a selection graph that includes component tags, descriptive properties, and contextual user information. This theoretical model enables the dynamic composition of user interfaces that adapt to different devices, user roles, and operational contexts.

Waste Management Theories and Approaches

The theoretical foundation for manure management incorporates circular economy principles, which aim to transform waste streams into valuable resources. In poultry operations, this involves technologies and processes that convert manure into fertilizer, energy, or other useful byproducts.

Probiotic methods represent a biological approach to odor control, leveraging beneficial microorganisms to suppress pathogenic bacteria and reduce ammonia formation. The use of absorptive materials such as charcoal and sawdust

provides a physical mechanism for moisture control, which indirectly reduces ammonia volatilization by creating less favorable conditions for bacterial decomposition (Yunos et al., 2024; Shao et al., 2015). Computational Fluid Dynamics (CFD) provides a theoretical framework for understanding and optimizing airflow within poultry houses. This approach models the complex interactions between ventilation systems, internal structures, and environmental conditions, enabling the design of layouts that minimize ammonia concentration while maintaining optimal temperature and humidity levels (Wu et al., 2024).

(***Review of Literature***- *discuss here the following: (a) related researches that have been conducted, state-of-the-art or current technologies from which the project will take off; (b) scientific basis/theoretical framework; and (c) results of related research conducted by the same Project Leader, if any. Not to exceed three (3) pages.*)

(6) OBJECTIVES

General Objective

To design and develop an IoT-based smart chicken coop system with automated manure management capabilities that effectively reduces health risks and environmental impacts associated with improper waste management in small-scale backyard poultry farming, while providing real-time monitoring and control of the coop environment specifically

- To design and construct a custom solar-powered automated manure collection coop system.
- To implement an environmental monitoring system that continuously tracks and records key parameters including temperature, humidity, ammonia concentration, and carbon dioxide levels for gas concentrations.
- To create a progressive web application interface that provides real-time data visualization, remote monitoring capabilities, and alert notifications.
- To develop an automated ventilation control system that responds to environmental sensor readings, maintaining optimal temperature levels within the chicken coop
- To conduct field testing and validation of the integrated system in at least three different small-scale backyard chicken coops in Iligan City over a period of three months, measuring key performance indicators including the farmer satisfaction.
- To perform a comparative analysis between traditional manure management practices and the proposed IoT-based system, documenting improvements in environmental impact, operational efficiency, and economic benefits through quantitative metrics.
- To develop a user manual and training program for small-scale poultry farmers that enables effective implementation, operation, and maintenance of the system with minimal technical expertise.

(***Objectives***- *statements of the general and specific purposes to address the problem areas of the project.*)

(7) METHODOLOGY

COMPONENT 1

Embedded Systems Development for Smart Chicken Coop Manure Management

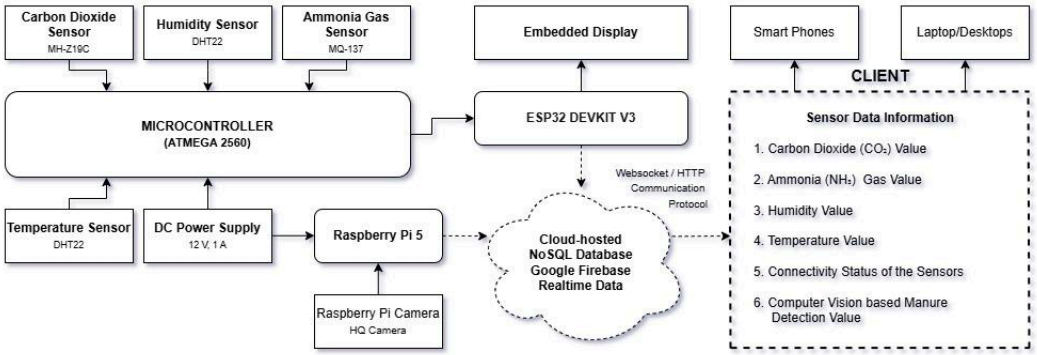


Figure 1: Block Diagram for Technical Architecture and IoT Protocol

The embedded systems development for the smart chicken coop manure management system in Figure 1 begins with the integration of specialized environmental sensors connected to an ATMEGA 2560 microcontroller as the primary data acquisition unit. This includes calibrating and programming the MH-Z19C carbon dioxide sensor, MQ-137 ammonia gas sensor, and DHT22 temperature/humidity sensors to collect measurements at certain sampling intervals.

- Firmware development focuses on implementing efficient sensor polling routines, signal conditioning algorithms, and error detection mechanisms to ensure data reliability even under challenging coop conditions. The ATMEGA microcontroller processes and consolidates raw sensor data before transmitting it to the ESP32 DEVKIT V3, which serves as the wireless communication bridge utilizing Websocket and HTTP protocols.

- The Raspberry Pi 5 subsystem is developed with emphasis on computer vision capabilities, Programming the Raspberry Pi with specialized image processing algorithms based on convolutional neural networks and implementing YOLO-V3 detection models

- Communication between the Raspberry Pi and microcontroller systems is established through serial interfaces. Power management systems are integrated throughout the embedded hardware, with the 12V DC power supply configured to provide stable power distribution with appropriate voltage regulation circuits for each component.

COMPONENT 2
IoT and Data Communication Implementation

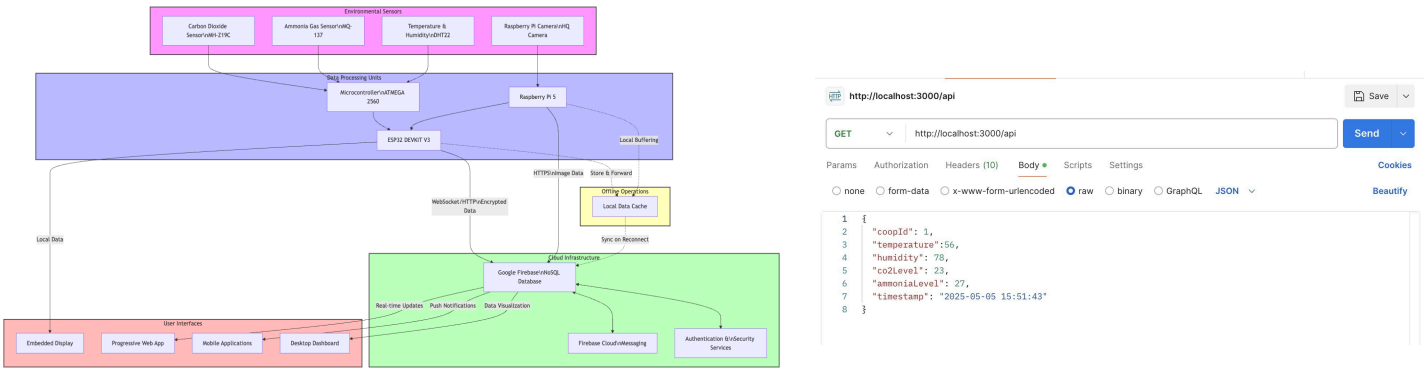


Figure 2: Data Communication and IoT System Protocol

The ESP32 DEVKIT V3 serves as the primary gateway seen in Figure 2, implementing secure WebSocket and HTTP protocols to transmit encrypted environmental and computer vision data to the Google Firebase real-time database. This bi-directional communication channel supports both data uploads and remote command execution.

- The system implements store-and-forward mechanisms with local data caching on both the ESP32 and Raspberry Pi, automatically synchronizing with the cloud when connectivity resumes.

- Authentication and encryption protocols protect all data transmissions, with device registration and certificate-based security measures preventing unauthorized access.

COMPONENT 3
Database Design and Development

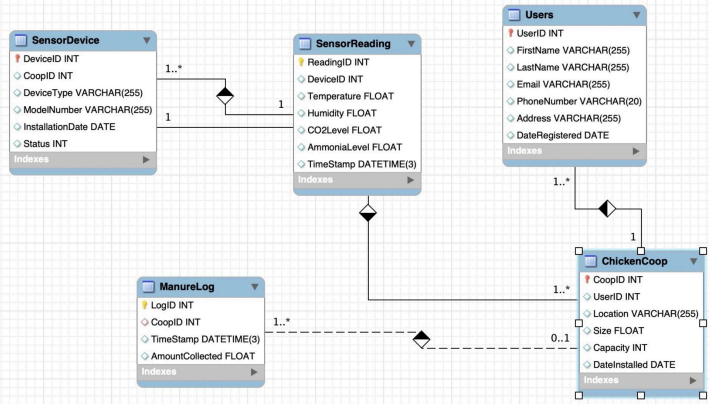


Figure 3: Entity Relationship Diagram (ERD)

The database design for the Smart Chicken Coop Monitoring System employs a normalized relational approach centered around five interconnected entities that efficiently model the system's domain as seen in Figure 3.

- The schema incorporates one-to-many relationships with appropriate cardinality constraints (1..* notation), enabling efficient queries across the entity hierarchy. Implementation considerations included strategic indexing for timestamp-based queries, particularly on the SensorReading table which will experience high data volume, and accommodations for time-series partitioning.

- The hierarchical structure flows from Users who own multiple ChickenCoops, each hosting multiple SensorDevices that generate numerous SensorReadings, while ManureLog entries are associated directly with specific coops. This design deliberately normalizes environmental data by separating device metadata (stored in SensorDevice) from actual readings (stored in SensorReading), preventing redundancy while maintaining referential integrity.

- Float data types were selected for sensor measurements (Temperature, Humidity, CO2Level, AmmoniaLevel) to accommodate decimal precision, while DATETIME(3) fields provide millisecond-level temporal resolution for accurate time-series analysis

COMPONENT 4
IoT Environmental Web Dashboard

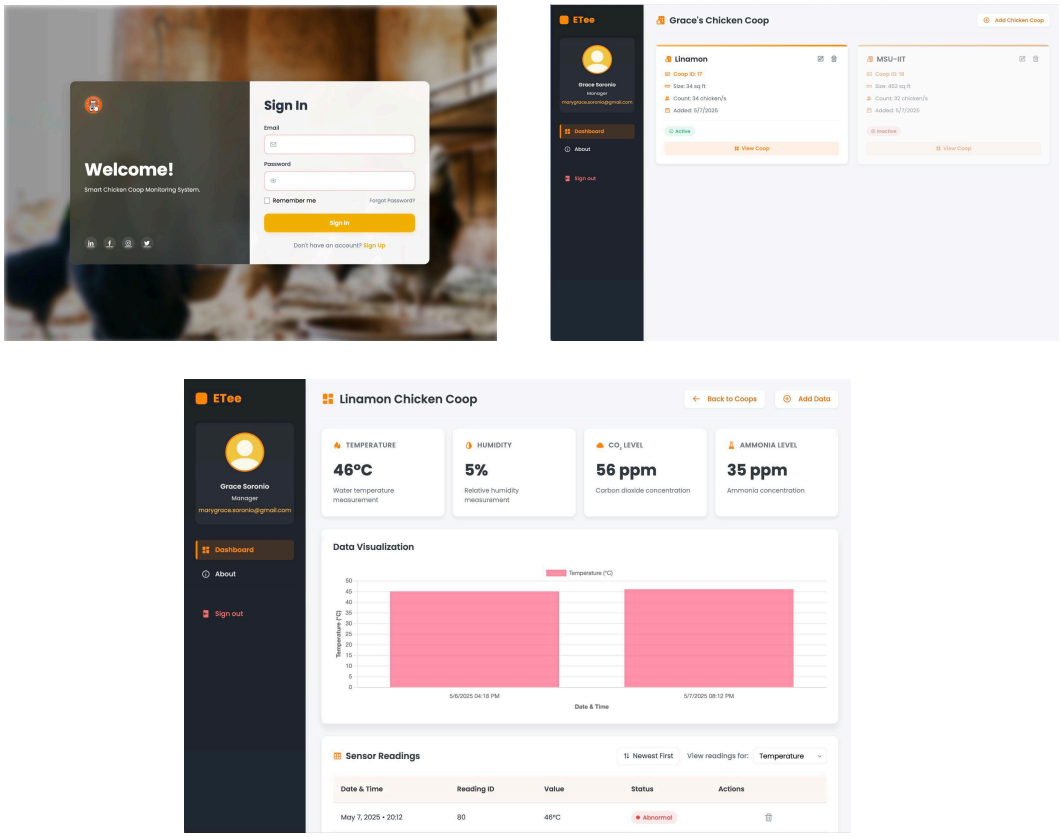


Figure 4: IoT Environmental Monitoring Dashboard

- The technical methodology for the Smart Chicken Coop IoT Dashboard establishes a robust architecture centered on a Node.js/Express backend infrastructure interfaced with a MySQL relational database. This backend implements a RESTful API pattern with standardized endpoints for sensor data ingestion, coop management, user authentication, and manure collection tracking.

- Authentication and security are implemented through a cookie token system with bcrypt password hashing and salting. The authentication flow begins with user credential verification against the database, followed by JWT token generation incorporating user role information for role-based access control (RBAC). Middleware functions intercept API requests to verify token validity and permission levels before allowing access to protected routes.

- For data visualization performance, the implementation employs canvas-based Chart.js rendering with data point throttling techniques that dynamically adjust resolution based on viewport size and date range.

COMPONENT 5
Structural Design of the Smart Chicken Coop

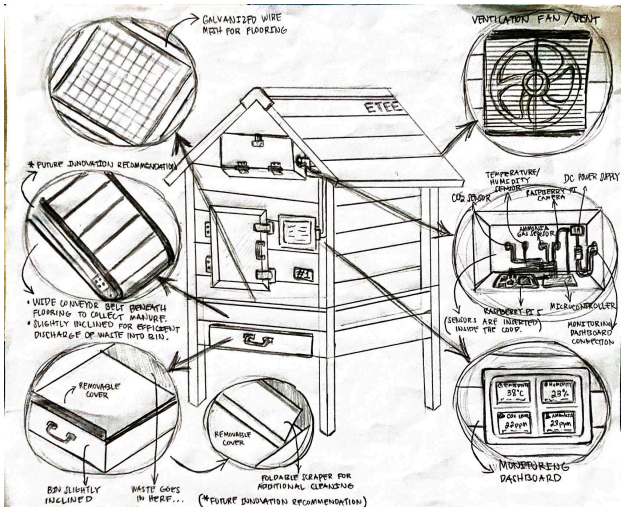


Figure 5: Schematic Sketch of the Structural Design Prototype

The methodology begins with constructing a raised chicken coop structure with integrated waste management capabilities as seen in Figure 5:

- A combination of galvanized wire mesh flooring allows manure to pass through while providing comfortable footing for chickens. The flooring system is designed with a slight inclination to facilitate the natural movement of waste material. Beneath this mesh flooring, a wide conveyor belt system is installed at a calculated angle to efficiently transport manure to a collection point.
- This conveyor system terminates at a removable waste collection bin featuring a secure cover to contain odors and prevent pest access. The coop design incorporates strategic ventilation points with an electric fan system to maintain optimal air circulation and temperature regulation.
- The waste management methodology employs both passive and active components. The primary passive system utilizes gravity and the inclined conveyor belt to continuously move waste toward the collection bin. The collection bin is designed with a slight internal inclination to concentrate waste at the deepest point, maximizing capacity before requiring emptying.
- Once the structure is complete, the environmental monitoring system is installed with careful calibration of each sensor against reference standards.. Initial testing involves running the system without chickens to establish baseline readings and verify mechanical operations. After introducing a small chicken population, the system undergoes continuous refinement through iterative calibration of sensor thresholds and adjustment of conveyor angles and speeds to optimize waste management efficiency

(***Methodology***- *discuss here the following: (a) variables or parameters to be measured and evaluated or analyzed; (b) treatments to be used and their layout; (c) experimental procedures and design; (d) statistical analysis; (e) evaluation method and observations to be made, strategies for implementation.*)

(8) ETHICAL/BIOSAFETY CLEARANCE

*(**Ethical clearance** from the agency's Institutional Ethics Review Committee (IERC) is required for researches involving the use of human subjects.*
***Biosafety clearance** is needed to ensure that all studies dealing with genetic engineering and pathogenic organisms in the Philippines are conducted under reasonably safe conditions.)*

(9) TARGET ACCOMPLISHMENTS AND EXPECTED OUTPUTS

(**Target Accomplishments**- measurable and positive results of completing project activities.)

(Expected Outputs- deliverables of the project based on the 6Ps and 2Is metrics (Publications, Patents/Intellectual Property, Products, People Services, Places and Partnerships, Policy, Social Impact and Economic Impact).
Publication- published aspect of the research, or the whole of it, in a scientific journal for peer review.

Patent/Intellectual Property- proprietary invention or scientific process for potential future profit.
Product- invention with a potential for commercialization.
People Services- people or groups of people, who receive technical knowledge and training.
Partnership- linkage forged because of the study.
Policy- science-based policy crafted and adopted by the government or academe as a result of the study.)
Social Impact
Economic Impact

(10) TARGET BENEFICIARIES

- **Local Government Unit - Iligan City**
- **Department of Agriculture**
- **Department of Health**
- **The Local Small Scale Chicken Farmers in Iligan City**

(**Target Beneficiaries**- Refers to groups/persons who will be positively affected by the conduct of the project.)

(11) LIMITATIONS OF THE PROJECT

There may be resistance from farmers in adopting new automated systems, and ongoing maintenance and monitoring of the IoT systems could pose challenges specifically:

- The environmental monitoring system relies on specific sensors (DHT22, MQ-137, MH-Z19C) which have inherent accuracy limitations, especially in high-humidity environments typical of chicken coops.
- The project is specifically designed for small-scale backyard chicken coops in Iligan City, with testing planned for only three locations, limiting broader generalizability to different climate zones or larger operations.

(**Limitations of the Project**- refers to restrictions or constraints in the conduct of the project.)

(12) LITERATURE CITED

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(**Literature Cited**- an alphabetical list of reference materials (books, journals and others) reviewed. Use a standard system for citation.)

(13) ESTIMATED BUDGETARY REQUIREMENT

Php _____

(Estimated Budgetary requirement - personal services (PS) and maintenance and other operating expenses (MOOE).)

(14) DETAILED LIB (Please fill up ORM-RP Form 02)

(15) Curriculum Vitae

Curriculum vitae - brief updated profile of the Project Leader. One (1) page only.

(16) OTHER ONGOING PROJECTS BEING HANDLED BY THE PROJECT LEADER: _____ (number)

(**Other Ongoing Projects Being Handled By The Project Leader**- list of ongoing projects being handled by the Project Leader funded internally or external funding agencies (DOST, CHED, USAID, etc,) and the accompanying responsibilities relevant to the project.)

Title of the Project	Funding Agency	Involvement in the Project

I hereby certify the truth of the foregoing. Any willful omission/false statement shall be a basis of disapproval and cancellation of the project.

(17)	SUBMITTED BY (Project Leader)	ENDORSED BY (Dept. Chairperson)
Signature		
Printed Name		
Date		
	ENDORSED BY (College Dean)	If PRISM Project, ENDORSED BY (PRISM Director)
Signature		
Printed Name		
Date		